

EDA Notebook

0. Setup Environment

```
In [1]: # DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.iprenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
12.9/12.9 MB 56.3 MB/s eta 0:00:00
4.9/4.9 MB 47.1 MB/s eta 0:00:00
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 which is incompatible.

Mounted at /content/gdrive

You can now save your data files in: /content/gdrive/MyDrive/36106/assignment/AT3/data

Student Information

```
In [ ]: group_name = ""
student_name = "Yi An Tsai"
student_id = "25532767"
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='student_name', value=student_name)
```

```
In [ ]: # Do not modify this code
print_tile(size="h1", key='student_id', value=student_id)
```

0. Python Packages

0.a Install Additional Packages

If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
In [ ]: # <Student to fill this section>
```

0.b Import Packages

```
In [ ]: # <Student to fill this section>
import pandas as pd
import altair as alt
```

B. Data Understanding

```
In [ ]: # Do not modify this code
try:
    df = pd.read_csv(at.folder_path / "unit_measure.csv")
except Exception as e:
    print(e)
```

B.1 Explore Dataset

```
In [ ]: df.head()
```

```
Out[ ]:
```

	unit_measure_code	name
0	ee0468c7-6417-489a-b233-da57ab966174	Boxes
1	7768b253-7430-402f-b90f-1e025c57459b	Bottle
2	1f485b4e-fc4d-47b2-ab57-af19b21eee40	Celsius
3	e87c6cb9-24ca-405a-996c-8f59ab7730fa	Canister
4	91fa7aa6-72aa-40a6-822f-8b80aa9f6376	Carton

```
In [ ]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38 entries, 0 to 37
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   unit_measure_code 38 non-null    object
1   name              38 non-null    object
dtypes: object(2)
memory usage: 740.0+ bytes
```

```
In [ ]: df.shape # 38 rows and 2 columns
```

```
Out[ ]: (38, 2)
```

```
In [ ]: len(df['unit_measure_code'].unique())
```

```
Out[ ]: 38
```

```
In [ ]: len(df['name'].unique())
```

```
Out[ ]: 38
```

```
In [ ]: df['unit_measure_code'].value_counts()
```

Out []:

unit_measure_code	count
ee0468c7-6417-489a-b233-da57ab966174	1
7768b253-7430-402f-b90f-1e025c57459b	1
1f485b4e-fc4d-47b2-ab57-af19b21eee40	1
e87c6cb9-24ca-405a-996c-8f59ab7730fa	1
91fa7aa6-72aa-40a6-822f-8b80aa9f6376	1
1f5f8ee7-58c4-4c4e-b8db-b7c1025b3400	1
b4c6422a-e170-4911-852f-38a7c19f5573	1
5c3d2f68-8573-4e53-8c4e-a72a5f92518e	1
e2e52698-ce31-4fb2-866f-bcffbdfd3147	1
7d2f574f-2c4b-45e5-bb2e-7c74c247492c	1
44f61043-62cb-4dea-a921-edf1df4b6a39	1
609a70c5-01f5-48f0-9939-4f04449e2018	1
0cc915c9-7473-4676-afff-2a4d4f023594	1
28bd7d2e-b75a-40dc-b62f-f360a9de2559	1
75fa117b-bb26-4752-8ec5-771acf4c50f6	1
8dbf3502-309f-4636-9f3e-1ae1cab3748b	1
cf068cf8-e6f9-4956-b31b-1d6eb91f3c5c	1
8c4f020f-f9c1-492b-8395-1372b7bda7a7	1
acbdd92d-1e1f-4ba8-a695-3f0c82d87759	1
7cca3216-2076-4b76-b746-8d4a33c94222	1
8dae8c54-6149-48bd-a0e1-9050f4ee82b6	1
d8954392-48a0-4cce-af48-15a83ab47274	1
ee700422-0d9b-48c5-ae33-d08b8b5034bf	1
57ee0c60-3085-4d9c-bae2-26022aa2d447	1
aea9d8c5-cf48-46e0-afa3-f36c6cf44c4e	1
c4797df8-be0b-4d3a-a6c4-0f1765f0a822	1
dd058911-cfcb-4c19-9df7-a8f40bbccf13	1
ee1d7851-244e-47db-8960-bced186cccad	1
8cb5f9fe-f273-42f3-96e4-1f93e26bb318	1
b0a224d6-adce-4062-b701-32d569ddaa05	1
3fbce844-926c-442f-87d9-9017094ed6b7	1
1316d404-c512-49da-bfb6-f6df816570ef	1
5c50e872-03a3-4aa1-8d0a-103cdf0fc00d	1
6ad6e4b7-e1c2-42f2-90f6-306f08b41623	1
2eb286e8-8050-4cf5-a260-fd2cedb6e1f4	1
5885bdb8-189e-4001-9f96-9cffd780b130	1
14266c5d-bb1d-4b58-9d8d-f13f6eb35863	1
a24c076e-f88e-44e9-811b-b4a3da40d463	1

dtype: int64

In []: df['name'].value_counts()

Out[]:

	count
name	
Boxes	1
Bottle	1
Celsius	1
Canister	1
Carton	1
Cubic meters	1
Cubic centimeter	1
Cubic decimeter	1
Centimeter	1
Square centimeter	1
Crate	1
Case	1
Container	1
Decimeter	1
Dozen	1
Each	1
Cubic foot	1
Gram	1
Gallon	1
Inch	1
Kilogram	1
Kilogram/cubic meter	1
Kilometer	1
Kiloton	1
Liter	1
US pound	1
Meter	1
Square meter	1
Cubic meter	1
Milligram	1
Milliliter	1
Millimeter	1
Ounces	1
Pack	1
Pallet	1
Piece	1
Percentage	1
Pint, US liquid	1

dtype: int64

```
In [ ]: dataset_insights = """
There are 38 rows and 2 columns in the dataset, without missing values.
This dataset includes 38 unique unit_measure_code and 38 unique name.
"""

In [ ]: # Do not modify this code
print_tile(size="h3", key='dataset_insights', value=dataset_insights)
```

B.2 Explore Feature of Interest \<put feature name here\>

```
In [ ]: feature_1_insights = """
There isn't any meaningful feature here, as these two columns simply represent the same information, the unique code and its corresponding name.
"""

In [ ]: # Do not modify this code
print_tile(size="h3", key='feature_1_insights', value=feature_1_insights)
```